

Testing Modified Inertia in Galaxy Clusters: An Anisotropy-Dependent Mass Normalization

Carl P. Zimmerman · Briar Creek Tech · 2026-07-01

Abstract

The residual missing-mass discrepancy of galaxy clusters — a factor of roughly two that survives even after the MOND boost — is shared by every realization of Milgrom’s law and is its central unsolved problem. Modified-inertia (MI) and modified-gravity (MG) formulations of MOND agree exactly on circular orbits but are known to diverge on non-circular ones; this divergence is Milgrom’s, established for disc galaxies through the global deep-MOND parameter (Milgrom 2012) and the general MG virial theorem (Milgrom 2014). This note applies that known divergence to *pressure-supported clusters* and extracts one definite, previously-unstated consequence. In modified inertia the dynamical-mass normalization η in the deep-MOND relation $G M a_0 = \eta \sigma^4$ is **system-dependent** — **it varies with the cluster’s orbital anisotropy** — **and it varies with a definite sign**: radial anisotropy *raises* η and *worsens* the residual, because Milgrom’s non-local inertial response is fixed by the orbit’s Fourier acceleration amplitude (pericentre-dominated), not by residence time. In modified gravity (AQUAL, QUMOND, AeST) and in Λ CDM the same normalization is *anisotropy-independent* — for MG by Milgrom’s virial theorem, for Λ CDM because the dynamical mass is simply the true mass. The predicted correlation “ η increases with radial-anisotropy β ” is therefore a signature no modified-gravity theory and no dark-matter model can produce, and it is automatically safe against the tight galaxy radial-acceleration relation because disc orbits are single-frequency. We reproduce the deep-MOND baseline on 9,830 eROSITA (eRASS1) clusters and state the test. We are explicit about scope: this **sharpens** Milgrom’s disc discriminator into a cluster test with a definite sign; it does **not** close the cluster residual — pure modified inertia floors *above* closure. The parent results are entirely Milgrom’s; the contribution here is the cluster-anisotropy realization and its sign.

1. The cluster residual and where MI and MG part ways

Milgrom’s law reproduces galaxy rotation curves with a single acceleration scale a_0 , but in galaxy clusters it leaves a residual: the dynamical mass still exceeds the MOND-boosted baryonic mass by a factor of order two (Sanders 1999; The & White 1988). This residual is *shared* — it appears identically in every relativistic and non-relativistic realization, and it is the standard, decades-old open problem of the paradigm (Famaey & McGaugh 2012).

MOND admits two distinct dynamical readings. In **modified gravity** (AQUAL, QUMOND, and the relativistic AeST theory of Skordis & Złótnik 2021) the field equation is altered and the modification is a property of the *potential*. In **modified inertia** the kinetic action is altered and the modification is a property of the *trajectory* (Milgrom 1994, 2022). On a circular orbit the two are indistinguishable — both reduce to the algebraic relation $g = v(g_N/a_0)$ that defines the radial-acceleration relation — which is why rotation curves cannot separate them. They separate only on *non-circular* orbits, and this is Milgrom’s own result: the global deep-MOND parameter Q distinguishes MI from MG on disc galaxies (Milgrom 2012), and the general MG virial theorem

fixes the MG mass–velocity normalization to be *universal* across systems (Milgrom 2014). A pressure-supported cluster is the extreme non-circular case: its member galaxies and gas move on eccentric, radial, and box orbits, not circles.

2. The modified-inertia cluster prediction: η slides, and the sign

In Milgrom’s modified-inertia formulation the inertial force on a body is a non-local functional of its whole trajectory. For a bound orbit decomposed into Fourier modes, the deep-MOND magnification is governed by the **acceleration amplitude spectrum** $\tilde{\omega}(\omega) \propto \sum_k \omega_k^2 |\tilde{r}_k|$ (Milgrom 2022, arXiv:2208.07073, Eqs. 20/27), not by how long the orbit dwells at any radius. This distinction is decisive and, we find, counter-intuitive. An eccentric orbit spends most of its *time* near apocentre, where the acceleration is lowest; one might expect that low-acceleration dwelling to deepen the MOND boost. It does not: the *amplitude* functional is dominated by the brief, high-acceleration pericentre passages. Every physical acceleration average that enters $\tilde{\omega}$ — the mean, the rms, even the harmonic mean — rises with eccentricity; only the pure apocentre value stays flat, and no term in Milgrom’s functional equals it.

Carried through to the cluster mass, this fixes both the *dependence* and its *sign*. Writing the deep-MOND pressure-supported relation as $G M a_0 = \eta \sigma^4$, the normalization η in modified inertia is not a constant: it slides with the orbital anisotropy β of the cluster, and **radial anisotropy raises it**. On the framework’s footing ($a_0 = c^2 \sqrt{\Lambda/32\pi} = 9.36 \times 10^{-11} \text{ m/s}^2$, its own convex interpolation), a direct evaluation over an eRASS1-calibrated deep-MOND ensemble gives the mean-field value $\eta \approx 2.15$ for isotropic orbits and $\eta \approx 2.8\text{--}3.0$ for radial anisotropy $\beta \approx 0.3\text{--}0.5$ — the residual grows, not shrinks. The mean-field (circular) value is a *floor*: no physical orbit population in the theory produces less residual than the circular case, so pure modified inertia cannot reach closure ($\eta = 1$) by orbital structure alone. That last point is a genuine sharpening of the standing cluster problem: it removes anisotropy as a candidate cure and shows the effect runs the wrong way.

3. The modified-gravity and dark-matter side: η is anisotropy-independent

The contrast is exact and, again, Milgrom’s. His general virial theorem for modified-gravity MOND (Milgrom 2014, PRD 89, 024016) shows that for *any* MG realization the mass–velocity-dispersion normalization is fixed by the theory’s basic tenets and is **the same value independent of the system’s internal structure** — in particular independent of its orbital anisotropy. AQUAL, QUMOND, and AeST all inherit this universality; they see only the mean gravitational field and cannot encode the trajectory. In Λ CDM the point is trivial: the dynamical mass is the actual gravitating mass, so the ratio carries no orbital-anisotropy dependence at all.

Therefore a measured correlation between the cluster mass normalization η and the orbital anisotropy β — with η rising toward radial orbits — is a signature that **no modified-gravity theory and no dark-matter model can produce**. It is the pressure-supported analogue of Milgrom’s disc Q-discriminant, with a specific predicted sign.

4. The test, and why it is galaxy-safe

Operationally: bin a cluster sample by orbital anisotropy (from resolved member kinematics, or from the joint X-ray-plus-lensing shape), and measure η in each bin. Modified inertia predicts a positive tilt $\eta(\beta)$; modified gravity, AeST, and Λ CDM predict a flat line. The test **cannot leak into the galaxy radial-acceleration relation**, because disc orbits are single-frequency: for a circular orbit Milgrom’s functional collapses to the algebraic $v(g_N/a_0)$ exactly (Milgrom 2022, Eq. 29), so an anisotropy-keyed effect is identically zero in rotationally-supported systems. The tight ~ 0.1 -dex RAR is untouched by construction — the discriminator lives only where support is by random motions.

A second, independent modified-inertia cluster signature points the same way: because MI inertia is trajectory-dependent, cluster members that are dynamically younger (on their first infall) carry a small excess velocity-dispersion relative to virialized members at the same cluster-centric radius, an effect that is exactly zero in the memoryless MG/AeST field theories. Both signatures share the same origin — the trajectory-dependence of inertia — and both are MG-impossible.

Neither is decisive with present data: current samples lack the resolved anisotropy measurements at the required precision, and the effect competes with the ordinary tidal and projection systematics of cluster kinematics. These are near-term, not in-hand, tests — realistically an eROSITA-plus-spectroscopy or ELT-era measurement.

5. Scope, credit, and what this does not claim

The dynamical divergence used here is Milgrom’s: the MI/MG distinction on non-circular orbits (Milgrom 2012), the MG virial universality (Milgrom 2014), and the non-local acceleration-amplitude kernel (Milgrom 2022) are all his. The contribution of this note is narrow and specific: the **cluster/pressure-supported anisotropy realization of that discriminator, carried through the kernel to a definite sign** (radial anisotropy worsens the residual; the mean-field value is a floor), together with its explicit galaxy-safety and its statement as a falsifiable $\eta(\beta)$ test.

We make no claim that this solves the cluster problem. It does the opposite in a useful way: it shows that pure modified inertia *cannot* close the residual through orbital structure, and that the residual continues to require a mass component — in the AeST realization, the ghost-condensate “dust,” whose amount is a free initial condition and is not predicted by the theory (Skordis & Złośnik 2021, who state plainly that this density “is not (classically) predicted”). The acceleration scale $a_0 = c^2\sqrt{(\Lambda/32\pi)}$ fixes the framework’s footing; the discriminator’s structure is largely independent of that value. This is offered in the spirit of a sharpening, not a discovery — a definite, testable sign attached to an existing discriminant.

References

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Reproducible baseline: eRASS1 (Bulbul et al. 2024) deep-MOND η on the framework footing $a_0 = c^2\sqrt{(\Lambda/32\pi)} = cH_\Lambda/Z$, $Z = \sqrt{(32\pi/3)}$. The author retracted all earlier “theory of everything” claims (2026-06-23); this note makes none. It is a scoped sharpening of Milgrom’s modified-inertia discriminator, credited as such, and explicitly does not solve the cluster mass discrepancy.